QUANTITATIVE ENERGY MODELING

POWER MARKETS / OPTION PRICING

Implied Volatilities & Forward Curve Calibration for illiquid markets

Inferring Swiss implied volatility surfaces from EEX benchmark data and calibrating a 3-factor HJM model for swing option valuation.

MARKET CONTEXT

The Swiss Power Volatility Challenge

Illiquidity in CH Exchange Options

Swiss (CH) power options lack direct exchange liquidity. The German (DE) market traded on EEX serves as the primary regional benchmark for pricing and calibration.

Two-Stage Calibration Framework

1. Infer CH ATM volatility surface via benchmark scaling.
2. Calibrate a 3-Factor log-normal HJM model to simulate forward curves for complex swing option valuation.

METHODOLOGY

Inferring CH Implied Volatility Surface

FRONT MONTH (M01)

$$IV_{CHM01} = IV_{DEM01} + DiffRealizedVol_{CH/DEM01}$$

Combines DE benchmark monthly volatility with historical realized differential between CH and DE spot returns.

FORWARD TENORS (M02+)

$$IV_{CHM_n} = IV_{CHM_{n-1}} \times \left(IV_{DE,ATMM_n} / IV_{DE,ATMM_{n-1}} \right)$$

Scales previous month's Swiss volatility using the ratio of consecutive DE At-The-Money (ATM) benchmark tenors.

LIMITATIONS

Shortfalls of Surface Inferencing

Uniform Strike Scale Assumption

Assumes implied volatility increases identically across all strikes. Creates significant risk of mis-evaluating the Samuelson effect on Out-of-the-Money (OTM) options.

Identical Monthly Co-Movement

Assumes Swiss monthly implied volatilities shift in strict proportion with German ones, ignoring local hydro supply and cross-border capacity bottlenecks.

MODELING FRAMEWORK

3-Factor Log-Normal HJM Framework

$$\frac{df_t^T}{f_t^T} = \frac{a}{T-t+d}dW_t^1 + \frac{b}{\sqrt{T-t+d}}dW_t^2 + cdW_t^3$$

FACTOR 1

Short-Term Decay

Fast exponential decay parameter **a**. Acts strongly on the prompt end of the curve.

FACTOR 2

Mid-Term Decay

Slow exponential decay parameter **b**. Governs mid-term maturity dynamics.

FACTOR 3

Parallel Shift

Constant parameter **c** across maturities. Captures market-wide shift factors.

CALIBRATION PIPELINE

PCA & Parameter Optimization Workflow

STEP 01

PCA on Historical Returns

Perform Principal Component Analysis on historical return series to construct the 3x3 correlation matrix of Brownian motions (W_1, W_2, W_3).

STEP 02

Volatility Matching

Fit parameters **{a, b, c, d}** by minimizing the sum of squared errors between market ATM implied volatility and analytical model volatility.

STEP 03

Granularity Transformation

Disaggregate simulated Monthly Power Forward Curves (MPFC) to Daily (DPFC) or Hourly (HPFC) curves using base HPFC coefficients.

RISK ANALYSIS

Forward Curve Simulation Shortfalls

Deterministic Volatility

Calibrated solely on ATM vol; ignores volatility skew and sudden volatility spikes.

Constant Correlations

Brownian motion correlation matrix is static, failing during market stress events.

Spot Volatility Blindness

Underestimates swing option value if exercised hourly due to missing intra-day dynamics.

Curve Smoothing

Parameter fitting smoothes out maturity-specific volatility spikes or troughs.

CONCLUSION

Key Takeaways for Swing Option Valuation

1. **EEX Benchmark Proxy:** DE ATM options provide a robust foundation for building CH volatility surfaces despite regional illiquidity.
2. **HJM Factor Structure:** The 3-factor model effectively captures curve term structure (short decay, mid decay, parallel shift) via historical PCA.
3. **Valuation Adjustments:** Spot volatility blindness means Monte Carlo swing option valuations should be adjusted upward for high-frequency exercise rights.